

ESPRIT DIRECTION-OF-ARRIVAL ESTIMATION IN THE PRESENCE OF SPATIALLY CORRELATED NOISE*

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Abstract. An algorithm is presented for ESPRIT (estimation of signal parameters via rotational invariance techniques) direction of arrival estimation when the expected value of the covariance of the measurement noise is different from a constant multiple of an identity matrix. The algorithm is based on a modification of the generalized singular value decomposition (GSVD) of two data matrices and requires only unitary transformations. The modification results in a significant simplification of the GSVD-based ESPRIT algorithm and produces more accurate solutions than prewhitening.

Key words. correlated noise, direction of arrival, generalized singular value decomposition

AMS subject classifications. 15A18, 15A23

1. Introduction. ESPRIT is a recently developed method for signal parameter estimation with applications in array signal processing such as direction of arrival estimation [7], [8]. The method is based on a sensor array consisting of two identical subarrays separated by a known displacement vector. It provides estimates of the signal parameters by exploiting the eigenstructure of the underlying rotational invariance. For other applications where similar structures arise, see [8]. We briefly describe the array geometry and signal model assumed in ESPRIT and show how the algorithm is formulated. Since the goal of this paper is to develop an efficient and numerically robust algorithm to estimate the directions of arrival according to the ESPRIT method, we will state the problem in a matrix-oriented language. For the detailed description and assumptions on which the formulation is based, see [7] and [8]. Throughout this paper, the superscript asterisk (*) of a matrix denotes the complex conjugate transpose of the matrix and i denotes the pure imaginary number $\sqrt{-1}$.

The array considered in ESPRIT consists of m sensor pairs. The sensor array is grouped into two subarrays, X and Y , that are assumed to be displaced by a known translation vector. We denote the number of sources by d , which is unknown, and assume the relation that $m \geq d$. Let $A \in C^{m \times d}$ be the unknown matrix with rank d , whose columns are the steering vectors associated with the sources, and let $\Phi = \text{diag}(e^{i\gamma_1}, \dots, e^{i\gamma_d}) \in C^{d \times d}$ be the unitary diagonal matrix that relates the measurements from subarray X to those from subarray Y . The sensor output $x(t)$, $y(t) \in C^{m \times 1}$ is modeled as

$$(1.1) \quad x(t) = As(t) + n_x(t), \quad y(t) = A\Phi s(t) + n_y(t),$$

where $s(t) \in C^{d \times 1}$ is an unknown vector of impinging signals and $n_x(t)$, $n_y(t) \in C^{m \times 1}$ are unknown noise vectors. Combining the two equations in (1.1), we have

$$(1.2) \quad z(t) \equiv \begin{bmatrix} x(t) \\ y(t) \end{bmatrix} = \begin{bmatrix} A \\ A\Phi \end{bmatrix} s(t) + n_z(t),$$

where

$$n_z(t) = \begin{bmatrix} n_x(t) \\ n_y(t) \end{bmatrix}.$$

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Assuming that the signals are uncorrelated with the noise and that the expected value of $s(t)s(t)^*$, $E(ss^*)$, is a positive definite covariance matrix R_{SS} , we have

$$E(zz^*) = \begin{bmatrix} A \\ A\Phi \end{bmatrix} R_{SS} \begin{bmatrix} A \\ A\Phi \end{bmatrix}^* + E(n_z n_z^*).$$

The problem is to find the unitary diagonal matrix Φ when we have the matrices $H_z \in C^{N_z \times 2m}$ and $H_n \in C^{N_n \times 2m}$ with full column rank that estimate $E(zz^*)$ and $E(n_z n_z^*)$, which is $E(zz^*) \approx H_z^* H_z$ and $E(n_z n_z^*) \approx \sigma^2 \Sigma = \sigma^2 H_n^* H_n$, with

$$(1.3) \quad H_z^* H_z \approx \begin{bmatrix} A \\ A\Phi \end{bmatrix} R_{SS} \begin{bmatrix} A \\ A\Phi \end{bmatrix}^* + \sigma^2 H_n^* H_n.$$

For notational convenience, we replace $\approx$ by $=$ in (1.3) in the following. Once the unitary diagonal matrix Φ is found, the directions of arrival can be derived from Φ by simple arithmetic. For details, see [7] and [8].

There have been many numerical algorithms developed for the ESPRIT direction of arrival methods [7], [8] where the noise is assumed to be white, i.e., the expected value of the covariance of the measurement noise has the form $E(n_z n_z^*) = \sigma^2 I$. In the more general case when $E(n_z n_z^*) = \sigma^2 \Sigma$, where $\Sigma \neq I$ is positive definite, prewhitening of the measurement noise using $\Sigma^{-1/2}$ avoids the generalized eigenvalue problem [8], [11]. However, when prewhitening is used, numerical difficulty can occur. To avoid the difficulty, a solution via a GSVD has been developed [13].

In this paper, we present an algorithm for the ESPRIT direction of arrival estimation when $E(n_z n_z^*) = \sigma^2 \Sigma$ where Σ is positive definite. The new algorithm produces numerically more accurate solutions than the prewhitening method and is more efficient than the existing GSVD-based method [13]. Our method is based on a special form of the GSVD, computation of which relies solely upon unitary transformations applied to the data matrices. The special form of the GSVD simplifies the computations in the ESPRIT by taking advantage of the structure of an intermediate matrix. Further advantages of the new form of the GSVD for ESPRIT will be discussed after the algorithm is introduced.

2. GSVD. The following theorem introduces the GSVD as was originally defined in Van Loan [12].

THEOREM 1. *Suppose that two matrices $H_z \in C^{N_z \times 2m}$ with $N_z \geq 2m$ and $H_n \in C^{N_n \times 2m}$ are given. Then there exist unitary matrices $U_z \in C^{N_z \times N_z}$ and $U_n \in C^{N_n \times N_n}$ and a nonsingular matrix $X \in C^{2m \times 2m}$ such that*

$$(2.1) \quad \begin{aligned} U_z^* H_z X &= D_z = \text{diag}(\alpha_1, \dots, \alpha_{2m}) \in C^{N_z \times 2m}, \\ U_n^* H_n X &= D_n = \text{diag}(\beta_1, \dots, \beta_p) \in C^{N_n \times 2m}, \end{aligned}$$

where $p = \min(N_n, 2m)$, $\alpha_i \geq 0$ for $1 \leq i \leq 2m$, and $\beta_i \geq 0$ for $1 \leq i \leq p$. $\square$

We will assume throughout this paper that $N_z \geq 2m$ and $N_n \geq 2m$, thus, $p = 2m$, since enough samples should be taken to provide good estimates for the directions of arrival. It is well known that a generalized eigenvalue problem for finding the scalars μ 's that satisfy $\det(H_z^* H_z - \mu H_n^* H_n) = 0$ can be solved by computing the GSVD of H_z and H_n since

$$\det(H_z^* H_z - \mu H_n^* H_n) = \det(D_z^* D_z - \mu D_n^* D_n) \det(X^{-*} X^{-1}).$$

This gives better numerical solutions since the explicit formation of the products $H_z^* H_z$ and $H_n^* H_n$ is avoided.

Although the GSVD can produce numerically more accurate solutions, computing the GSVD is a difficult task when the matrices are ill conditioned [4]–[6]. If the non-singular matrix X from the GSVD is required, then we can expect numerical difficulties when the data matrices H_z and H_n are ill conditioned. Paige and Saunders [5] suggest an alternative form of the GSVD that has better numerical properties. Paige [4] proposed an algorithm that computes this alternative form for the GSVD. We define yet another form of the GSVD in Theorem 2 upon which the new algorithm for the ESPRIT relies. The new GSVD of (2.2) is identical to the GSVD introduced by Paige and Saunders except that in [5], the matrix L is upper triangular. It will become clear that it is critical to have the lower triangular matrix L for simplifying the computation of the ESPRIT algorithm.

THEOREM 2. *Suppose that two matrices $H_z \in C^{N_z \times 2m}$ and $H_n \in C^{N_n \times 2m}$ with $N_z \geq 2m$ and $N_n \geq 2m$ are given. Then there exist unitary matrices $U_z \in C^{N_z \times N_z}$, $U_n \in C^{N_n \times N_n}$, and $Q \in C^{2m \times 2m}$, and lower triangular matrices $L_z \in C^{N_z \times 2m}$ and $L_n \in C^{N_n \times 2m}$, such that*

$$(2.2) \quad U_z^* H_z Q = L_z \quad \text{and} \quad U_n^* H_n Q = L_n,$$

where $L_z = D_z L$ and $L_n = D_n L$ for a nonsingular lower triangular matrix $L \in C^{2m \times 2m}$ and diagonal matrices $D_z = \text{diag}(\alpha_1, \dots, \alpha_{2m}) \in C^{N_z \times 2m}$ and $D_n = \text{diag}(\beta_1, \dots, \beta_{2m}) \in C^{N_n \times 2m}$ with $\alpha_i \geq 0$ and $\beta_i \geq 0$ for $1 \leq i \leq 2m$. Moreover, if each of H_z and H_n has full column rank, then the elements in D_z and D_n can be ordered so that

$$(2.3) \quad \frac{\alpha_1}{\beta_1} \geq \frac{\alpha_2}{\beta_2} \geq \dots \geq \frac{\alpha_{2m}}{\beta_{2m}}$$

is satisfied.

Proof. Let

$$(2.4) \quad U_z^* H_z X = D_z \quad \text{and} \quad U_n^* H_n X = D_n$$

be the GSVD of H_z and H_n according to Theorem 1. Consider the decomposition of X into a unitary matrix $Q \in C^{2m \times 2m}$ and a lower triangular matrix $L^{-1} \in C^{2m \times 2m}$: $X = QL^{-1}$. Then we have $U_z^* H_z Q = D_z L$ and $U_n^* H_n Q = D_n L$. The second part is obvious since the diagonal elements of D_z and D_n in the decomposition (2.4) can be ordered to satisfy the condition (2.3) via a permutation P , then the above process can be applied to XP . $\square$

Although Theorem 2 introduces a modified form for the GSVD that gives a numerically more robust and efficient algorithm for ESPRIT, our algorithm to compute the GSVD of (2.2) would not follow the procedure given in the proof. A better way to compute the new GSVD is given in Algorithm GSVD_L in § 3, which is essentially the same as the algorithm due to Paige [4], except that a matrix W in each step is chosen to lower triangularize the 2×2 submatrices, whereas Paige upper triangularizes the submatrices.

THEOREM 3. *Suppose that $H_z \in C^{N_z \times 2m}$ and $H_n \in C^{N_n \times 2m}$ have full column rank and that they have the GSVD as in (2.2) and D_z and D_n are ordered to satisfy (2.3). Also, assume that $H_z^* H_z = [A_\Phi] R_{SS} [A_\Phi]^* + \sigma^2 H_n^* H_n$, where $A \in C^{m \times d}$ has rank d , $\Phi \in C^{d \times d}$ is unitary and diagonal, and R_{SS} is symmetric positive definite. Then*

$$(2.5) \quad \frac{\alpha_1}{\beta_1} \geq \dots \geq \frac{\alpha_d}{\beta_d} \geq \frac{\alpha_{d+1}}{\beta_{d+1}} = \dots = \frac{\alpha_{2m}}{\beta_{2m}} = \sigma$$

and

$$(2.6) \quad \begin{bmatrix} A \\ A\Phi \end{bmatrix} R_{SS} \begin{bmatrix} A \\ A\Phi \end{bmatrix}^* = Q_1 L_{11}^* \Sigma L_{11} Q_1^*,$$

where $Q = [Q_1 \ Q_2]$ with $Q_1 \in C^{2m \times d}$, $Q_2 \in C^{2m \times (2m-d)}$,

$$L = \begin{bmatrix} L_{11} & 0 \\ L_{21} & L_{22} \end{bmatrix}$$

with $L_{11} \in C^{d \times d}$, and $\Sigma = \text{diag}(\alpha_1^2 - \sigma^2 \beta_1^2, \alpha_2^2 - \sigma^2 \beta_2^2, \dots, \alpha_d^2 - \sigma^2 \beta_d^2) \in R^{d \times d}$.

Proof. For the proof of the first part that involves (2.5), see [13].

$$\begin{aligned} \begin{bmatrix} A \\ A\Phi \end{bmatrix} R_{SS} \begin{bmatrix} A \\ A\Phi \end{bmatrix}^* &= H_z^* H_z - \sigma^2 H_n^* H_n = QL^*(D_z^* D_z - \sigma^2 D_n^* D_n)LQ^* \\ &= [Q_1 \ Q_2] \begin{bmatrix} L_{11}^* & L_{21}^* \\ 0 & L_{22}^* \end{bmatrix} \begin{bmatrix} \Sigma & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} L_{11} & 0 \\ L_{21} & L_{22} \end{bmatrix} \begin{bmatrix} Q_1^* \\ Q_2^* \end{bmatrix} \\ &= Q_1 L_{11}^* \Sigma L_{11} Q_1^*. \quad \square \end{aligned}$$

Dividing the matrix $Q_1 \in C^{2m \times d}$ further into two parts

$$Q_1 = \begin{bmatrix} Q_x \\ Q_y \end{bmatrix},$$

where $Q_x \in C^{m \times d}$ and $Q_y \in C^{m \times d}$, we have

$$(2.7) \quad \begin{bmatrix} A \\ A\Phi \end{bmatrix} R_{SS} \begin{bmatrix} A \\ A\Phi \end{bmatrix}^* = \begin{bmatrix} Q_x \\ Q_y \end{bmatrix} L_{11}^* \Sigma L_{11} \begin{bmatrix} Q_x \\ Q_y \end{bmatrix}^*.$$

Accordingly,

$$AR_{SS}A^* - \lambda AR_{SS}\Phi^*A^* = AR_{SS}(I - \lambda\Phi^*)A^* = Q_x L_{11}^* \Sigma L_{11} (Q_x^* - \lambda Q_y^*),$$

and $\text{rank}(AR_{SS}(I - \lambda\Phi^*)A^*) < d$ if and only if λ is the same as one of the diagonal elements of Φ . That is, to find Φ , we have only to find $\lambda_1, \dots, \lambda_d$ that make

$$(2.8) \quad \text{rank}(Q_x L_{11}^* \Sigma L_{11} (Q_x^* - \lambda Q_y^*)) < d, \quad \text{i.e., } \text{rank}(Q_x^* - \lambda Q_y^*) < d.$$

The importance of the lower triangular structure of the matrix L in the GSVD of (2.2) is clear from Theorem 3, since only the upper left $d \times d$ submatrix L_{11} of L is required in representing the matrix $\begin{bmatrix} A \\ A\Phi \end{bmatrix} R_{SS} \begin{bmatrix} A \\ A\Phi \end{bmatrix}^*$ as shown in (2.7). In fact, none of the elements from L is needed: the lower triangular structure of the matrix L resulted in requiring only Q_x and Q_y for finding the matrix Φ as shown in (2.8). The difficult problem that remains is that the eigenvalues of a rectangular matrix pencil $Q_x^* - \lambda Q_y^* \in C^{d \times m}$ must be computed. We can apply one of two previously developed approaches [8], [13] for reducing the problem to an eigenproblem of a square pencil. From the relation in (2.7),

$$(2.9) \quad \text{null}(Q_x^*) = \text{null}(Q_y^*) = \text{null}(A^*).$$

Let a matrix $J_1 \in C^{m \times (m-d)}$ satisfy $\text{span}(J_1) = \text{null}(Q_x^*)$ and be expanded to a nonsingular matrix $J = [J_1 \ J_2] \in C^{m \times m}$. Then

$$(2.10)$$

$$\text{rank}(Q_x^* - \lambda Q_y^*) = \text{rank}((Q_x^* - \lambda Q_y^*)J) < d, \quad \text{i.e., } \text{rank}(Q_x^* J_2 - \lambda Q_y^* J_2) < d,$$

and the values λ are the eigenvalues of a $d \times d$ square pencil. The matrix J can be obtained from the QR decomposition of Q_x [13].

We can also reduce the rectangular matrix pencil into a square pencil by using the total least squares (TLS) technique [2] as in TLS-ESPRIT [8]. Since

$$(2.11) \quad \text{Range}(Q_x) = \text{Range}(Q_y) = \text{Range}(A)$$

from (2.7), there must be a unique nonsingular matrix $T \in C^{d \times d}$ such that

$$(2.12) \quad Q_x = Q_y T.$$

Since

$$\text{rank}(Q_x^* - \lambda Q_y^*) = \text{rank}(T^* Q_y^* - \lambda Q_y^*) < d, \quad \text{i.e., } \text{rank}(T^* - \lambda I) < d,$$

we have only to find the eigenvalues of T^* . In practice, neither of the relations (2.9) or (2.11) can hold exactly in finite precision. We estimate the matrix T using the TLS idea as in TLS-ESPRIT, assuming that Q_x and Q_y are equally noisy [2], [8]. The algorithm is summarized in § 3.

3. New algorithm. In general, the matrices $H_z \in C^{N_z \times 2m}$ and $H_n \in C^{N_n \times 2m}$ in ESPRIT satisfy the relations that $N_z \gg 2m$ and $N_n \gg 2m$. The initial factorization of each matrix into a unitary matrix and a lower triangular form (we will call it a QL factorization),

$$H_z = Q_z \begin{bmatrix} L_z \\ 0 \end{bmatrix} \quad \text{and} \quad H_n = Q_n \begin{bmatrix} L_n \\ 0 \end{bmatrix},$$

reduces the computational costs for the GSVD substantially. If the GSVD of L_z and L_n is

$$(3.1) \quad U_1^* L_z Q = L_1, \quad U_2^* L_n Q = L_2,$$

where $L_1 = D_1 L$ and $L_2 = D_2 L$, then the GSVD of H_z and H_n can be obtained as

$$(3.2) \quad \left(Q_z \begin{bmatrix} U_1 & 0 \\ 0 & I \end{bmatrix} \right)^* H_z Q = \begin{bmatrix} L_1 \\ 0 \end{bmatrix}, \quad \left(Q_n \begin{bmatrix} U_2 & 0 \\ 0 & I \end{bmatrix} \right)^* H_n Q = \begin{bmatrix} L_2 \\ 0 \end{bmatrix}.$$

Note that the same unitary matrix Q appears in both (3.1) and (3.2), which is the only unitary matrix we need from the decomposition for our ESPRIT algorithm. Thus the problem is reduced to computing the GSVD of two lower triangular matrices $L_z, L_n \in C^{2m \times 2m}$, and there is no need to compute Q_z, Q_n, U_1 , or U_2 . The Paige algorithm [4] for computing the GSVD of two real matrices is similar to the Jacobi algorithm for computing the SVD. If we start with a triangular matrix and use cyclic-by-rows ordering, the matrix will alternate between upper and lower triangular forms after each sweep. Since the lower triangular structure of L is required in our algorithm, we can finish the iterations after an even number of sweeps when we start with two lower triangular matrices L_z and L_n . For parallel implementation, we can also use the odd-even ordering that preserves the original triangular structure. For details, see [1] and [4].

The algorithm for computing our GSVD is summarized in Algorithm GSVD_L. For a matrix X , X_{pq} denotes the 2×2 submatrix of X :

$$\begin{bmatrix} x_{pp} & x_{pq} \\ x_{qp} & x_{qq} \end{bmatrix}.$$

Similarly, for a 2×2 matrix Y , $Y(p, q)$ denotes a matrix of proper dimension that is like an identity matrix, except that its 2×2 submatrix in the (p, q) plane is Y .

ALGORITHM GSVD_L ($H_z \in C^{N_z \times 2m}$, $H_n \in C^{N_n \times 2m}$)

0. $Q := I_{2m}$.
1. Compute the QL decomposition of H_z and H_n :

$$H_z = Q_z \begin{bmatrix} A \\ 0 \end{bmatrix} \quad \text{and} \quad H_n = Q_n \begin{bmatrix} B \\ 0 \end{bmatrix}.$$

2. Repeat until convergence.
3. Iterate for one sweep using odd-even ordering.
/* assume that the current index pair is (p, q) .
- 3.1. Determine 2×2 unitary matrices U , V , and J
so that $U^* A_{pq} J$ and $V^* B_{pq} J$ are lower triangular and
the second row of $U^* A_{pq} J$ is parallel to the second row of $V^* B_{pq} J$.
- 3.2. { update the matrices }
 $A := U^*(p, q) A J(p, q)$,
 $B := V^*(p, q) B J(p, q)$,
 $Q := Q J(p, q)$.

The algorithm is terminated when each row of A is parallel to the corresponding row of B . Under the assumption that H_z and H_n have full column rank, the computed generalized singular values α_i / β_i can be obtained from a_{ii} / b_{ii} . For our application, none of the left unitary transformations need to be saved. Thus, the input matrices $H_z \in C^{N_z \times 2m}$ and $H_n \in C^{N_n \times 2m}$ can be overwritten by the lower triangular matrices $L_z \in C^{2m \times 2m}$ and $L_n \in C^{2m \times 2m}$. The discussions from § 2 are summarized in the following algorithm, using the TLS approximation.

ALGORITHM ESPRIT_GSVD_L

1. Form the matrices H_z and H_n from the available measurements.
2. Compute the GSVD of H_z and H_n

$$U_z^* H_z Q = L_z \quad \text{and} \quad U_n^* H_n Q = L_n$$

according to Algorithm GSVD_L.

3. Estimate the number of sources d from L_z and L_n .
Let

$$\begin{bmatrix} Q_x \\ Q_y \end{bmatrix}$$

be the first d columns of Q .

4. Compute the SVD of

$$[Q_x \quad Q_y] = W D \begin{bmatrix} V_{11} & V_{12} \\ V_{21} & V_{22} \end{bmatrix}^*.$$

5. Compute the eigenvalues of $-V_{12} V_{22}^{-1}$.

Two existing algorithms, one based on the GSVD of form (2.2) due to Van Loan and the other based on the TLS-ESPRIT with prewhitening, are summarized in algorithms ESPRIT_GSVD and ESPRIT_PREWHITENING, respectively. For the definition of the CS decomposition used in Algorithm ESPRIT_GSVD, see Golub and Van Loan [3].

ALGORITHM ESPRIT_GSVD

1. Form the matrices H_z and H_n from the available measurements.
2. Compute the QR decomposition

$$\begin{bmatrix} H_z \\ H_n \end{bmatrix} = \begin{bmatrix} Q_1 \\ Q_2 \end{bmatrix} R,$$

where $Q_1 \in C^{N_z \times 2m}$, $Q_2 \in C^{N_n \times 2m}$, and $R \in C^{2m \times 2m}$.

3. Compute the CS decomposition of Q_1 and Q_2 :

$$U_1^* Q_1 Q = C \quad \text{and} \quad U_2^* Q_2 Q = S.$$

4. Estimate the number of sources d from C and S .

Let

$$\begin{bmatrix} Q_x \\ Q_y \end{bmatrix}$$

be the first d columns of $R^* Q$.

5. Compute the SVD of

$$[Q_x \quad Q_y] = W D \begin{bmatrix} V_{11} & V_{12} \\ V_{21} & V_{22} \end{bmatrix}^*.$$

6. Compute the eigenvalues of $-V_{12} V_{22}^{-1}$.

Note that the matrices Q_1 , Q_2 , and R from step 2 need to be saved for steps 3 and 4. Thus the storage requirement of Algorithm ESPRIT_GSVD is larger than that of Algorithm ESPRIT_GSVD_L. Assuming that the complexity for computing the GSVD in ESPRIT_GSVD_L and the complexity for the CS decomposition in ESPRIT_GSVD are about the same, ESPRIT_GSVD is more costly and the control is more complicated than that of ESPRIT_GSVD_L since Q_1 , Q_2 , and $R^* Q$ should be computed.

ALGORITHM ESPRIT_PREWHITENING

1. Form the matrices H_z and H_n from the available measurements.
Measure Σ such that $E(n_z n_z^*) = \sigma^2 \Sigma$ (or assume that Σ is known).
 2. Compute $\Sigma^{-1/2}$.
 3. Compute the SVD of $H_z \Sigma^{-1/2}$: $H_z \Sigma^{-1/2} = U D Q^*$.
 4. Estimate the number of sources d from D .
- Let

$$\begin{bmatrix} Q_x \\ Q_y \end{bmatrix}$$

be the first d columns of $\Sigma^{1/2} Q$.

5. Compute the SVD of

$$[Q_x \quad Q_y] = W D \begin{bmatrix} V_{11} & V_{12} \\ V_{21} & V_{22} \end{bmatrix}^*.$$

6. Compute the eigenvalues of $-V_{12} V_{22}^{-1}$.

All three algorithms were coded in MATLAB on a SUN 3/80 with the machine precision $\varepsilon = 2.2204 E - 16$. Both algorithms ESPRIT_GSVD_L and ESPRIT_GSVD

produced solutions that are essentially the same in accuracy. When the condition number of the noise matrix H_n is large, prewhitening suffered from numerical difficulties. For example, when the condition number of $H_n \approx 1/\sqrt{\epsilon}$, with the directions of arrival 14° and 15° , and signal-to-noise ratio 10, while two algorithms based on the GSVD produced answers accurate to the ninth digit, prewhitening failed (the eigenvalues produced in step 6 have absolute values far from unit or the algorithm broke down without completion) 25 times out of 30 tests.

Remarks. We have presented a new algorithm for the ESPRIT direction of arrival estimation in the presence of noise of which the expected value of the covariance is $\sigma^2 \Sigma$, where Σ is not necessarily an identity matrix. The new algorithm has been shown to be more efficient than the existing GSVD-based algorithm in terms of storage and complexity. Also, it produces more accurate solutions than the prewhitening method. The fact that only the unitary matrix Q is needed would make the systolic array for the new algorithm simple. The implementation of the new algorithm on a two-dimensional mesh-connected array should be straightforward from the results presented in [6].

One difficulty in the parallel implementation of the proposed algorithm is that the generalized singular values need to be ordered in decreasing order. We have modified the Paige algorithm so that the generalized singular values of the 2×2 submatrices in each step are ordered. More research is needed regarding the relation between the ordering of the singular values and the convergence.

Recently, Stewart [9], [10] introduced new decompositions of a matrix based on two-sided unitary transformations, and they are called the URV and ULV decompositions. These can be generalized for the triangularization of a matrix pair analogous to the GSVD. The generalized URV and ULV decompositions have many potential applications, especially where the solutions should adapt to changing statistics. Note that (2.2) can be used as a starting point for such generalization of the URV and ULV decompositions. Rank estimation and recursive updating will need to be incorporated to fully realize the generalized URV and generalized ULV decompositions. The advantage of this approach is that the solution can be recursively updated in an efficient way, which will lead to an adaptive ESPRIT algorithm.

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